

Multi-channel consistency of the matter-bounce non-Gaussianity: pulsar-timing spectral slope, primordial black holes, and the SPHEREx-era bispectrum at $f_{\text{NL}}^{\text{local}} = -35/16$

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A dust-dominated contracting phase preceding a nonsingular bounce predicts a parameter-free local non-Gaussianity of the primordial curvature perturbation. The printed value of Cai *et al.* is $f_{\text{NL}}^{\text{local}} = -35/8$; a permutation-convention audit following Li *et al.* gives exactly half of it, $f_{\text{NL}}^{\text{local}} = -35/16 = -2.1875$, and that factor of two is not yet settled by an independent derivation. We ask what the halved value does to three observational channels, and we state each channel's result at exactly the strength its evidence supports. (i) *Pulsar timing*. Refitting the NANOGrav 15-yr Hellings–Downs-correlated Ceffyl free-spectrum posteriors, we obtain a characteristic-strain spectral index $\gamma = 2.567 \pm 0.382$, consistent at 1.14σ with the matter-bounce scalar-induced gravitational-wave prediction $\gamma = 3$ ($\Omega_{\text{GW}} \propto f^2$), and disfavouring the supermassive-black-hole binary value $\gamma = 13/3$ by a Savage–Dickey factor $\log_{10} B_{\text{MB/SMBHB}} = +3.85$. We emphasise that $\Omega_{\text{GW}} \propto f^2$ is the causality-limited infrared slope of *any* scalar-induced background, so this channel is a consistency check and not a discriminator, and that it is insensitive to the value of f_{NL} . A scale-invariant *primordial* tensor background ($n_{\text{T}} = 0$, $\gamma = 5$) is excluded at 6.37σ by the same posterior. (ii) *Primordial black holes*. Within a Press–Schechter treatment with the local quadratic map, negative f_{NL} imposes an absolute ceiling $\zeta_{\text{max}} \simeq -5/(12f_{\text{NL}})$ on the curvature perturbation, which therefore *doubles* from 0.0952 at $-35/8$ to 0.1905 at $-35/16$. At an amplitude calibrated so that the Gaussian case saturates $f_{\text{PBH}} = 1$, the halved value yields $f_{\text{PBH}} = 7.3 \times 10^{-3}$ against 3.8×10^{-6} at $-35/8$ – a suppression weaker by three orders of magnitude, placing $-35/16$ inside rather than far below the $10^{-3} \leq f_{\text{PBH}} \leq 1$ band reported for negative- f_{NL} bounce scenarios. (iii) *Large-scale structure*. The prediction is consistent with the DESI DR1 constraint $f_{\text{NL}}^{\text{local}} = -3.6_{-9.1}^{+9.0}$ at 0.16σ , and is the only channel sensitive to the value of f_{NL} : SPHEREx reaches 3.13σ (2.63σ after the bounce-to-local shape projection $r = 0.84$) at its bispectrum-only forecast $\sigma_{f_{\text{NL}}} = 0.7$, and 4.38σ (3.68σ) at its combined target $\sigma_{f_{\text{NL}}} = 0.5$. The three channels are mutually consistent. The operative uncertainty is internal rather than observational: only the bispectrum channel depends on which of $-35/16$ or $-35/8$ is correct, and it depends on it strongly.

I. INTRODUCTION

The matter-bounce scenario – a dust-dominated contracting phase matched through a nonsingular bounce onto the standard expanding history – predicts a scale-invariant curvature power spectrum together with a definite, parameter-free local bispectrum amplitude [1–3]. Section II fixes the value we test; Secs. III–V take it through three channels; Sec. VI states what is and is not established.

II. THE PARAMETER UNDER TEST

Cai *et al.* [1] print a squeezed-limit amplitude $f_{\text{NL}}^{\text{local}} = -35/8 = -4.375$ for the matter bounce. Li *et al.* [4], whose no-go analysis extends that of Quintin *et al.* [3], obtain exactly one half of it,

$$f_{\text{NL}}^{\text{local}} = -\frac{35}{16} = -2.1875. \quad (1)$$

The discrepancy traces to permutation counting in the in-commutator and to a $+(99/128) \sum_i k_i^3$ term carried by the Cai shape polynomial, whose removal halves the bare squeezed limit exactly. We adopt Eq. (1) throughout and carry $-35/8$ in parallel wherever a channel distinguishes them.

Scope statement.—The factor of two has *not* been settled by an independent second-method derivation (for example a Salopek–Bond gradient expansion or a δN computation). We therefore never state a channel result at $-35/16$ alone where the two values differ materially.

III. CHANNEL I: THE PULSAR-TIMING SPECTRAL SLOPE

A. Spectral conventions

With the characteristic strain parameterised as $h_c(f) \propto f^{(3-\gamma)/2}$ and $\Omega_{\text{GW}} = (2\pi^2/3H_0^2) f^2 h_c^2(f)$, the energy density scales as

$$\Omega_{\text{GW}}(f) \propto f^{5-\gamma}. \quad (2)$$

Hence $\gamma = 13/3$ for gravitational-wave-driven supermassive-black-hole binary inspirals ($\Omega_{\text{GW}} \propto f^{2/3}$),

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TABLE I. Channel I. Savage–Dickey factors and Gaussian z -distances from the posterior of Eq. (3). All values reproduced from the committed chain (`pta_gamma_reproduce.py`); differences from the archived record are $\leq 3 \times 10^{-15}$.

Hypothesis	γ_*	Ω_{GW} slope	$B_{\gamma_*/\text{free}}$	z
Matter bounce (induced)	3	f^2	3.23	1.14σ
SMBHB inspiral	13/3	$f^{2/3}$	4.52×10^{-4}	4.63σ
Primordial tensors	5	f^0	1.86×10^{-24}	6.37σ

$\gamma = 5$ for a scale-invariant primordial tensor background ($n_{\text{T}} = 0$, $\Omega_{\text{GW}} \propto f^0$), and $\gamma = 3$ for $\Omega_{\text{GW}} \propto f^2$, the universal infrared scaling of the scalar-induced background of a nonsingular matter bounce [5]. We stress that $\gamma = 3$ is *not* the $n_{\text{T}} = 0$ case; the two are separated by two units of slope.

B. Posterior and Bayes factors

We fit the NANOGrav 15-yr Hellings–Downs-correlated free-spectrum Ceffyl kernel density posteriors [6, 7] (30 frequency bins, $T_{\text{obs}} = 16.03$ yr) with a power law ($\gamma, \log_{10} A$) under priors $\gamma \sim U[0, 7]$, $\log_{10} A \sim U[-18, -11]$, using 32 walkers and 10^4 production steps after 2500 burn-in (3.2×10^5 samples, ESS = 5.5×10^3). The marginal is

$$\gamma = 2.567 \pm 0.382, \quad \log_{10} A = -14.03 \pm 0.38, \quad (3)$$

with median 2.591 and 68% interval [2.304, 2.882].

Savage–Dickey density ratios for the nested point hypotheses $\gamma = \gamma_*$ against free γ , $B_{\gamma_*/\text{free}} = p(\gamma_*|d)/\pi(\gamma_*)$ with $\pi = 1/7$, are collected in Table I.

The matter-bounce-to-SMBHB ratio is $B_{\text{MB}/\text{SMBHB}} = 7.14 \times 10^3$, i.e. $\log_{10} B = +3.85$. Narrowing the γ prior from $U[0, 7]$ to $U[2, 5]$ moves $B_{3/\text{free}}$ from 3.23 to 1.47 while leaving $B_{\text{MB}/\text{SMBHB}}$ stable in the range 7.1–8.7 $\times 10^3$; the nested factor is prior-sensitive, the model ratio is not.

C. What this channel does and does not establish

The measured slope is consistent with the matter-bounce induced-GW prediction and strongly disfavours both the SMBHB expectation and a scale-invariant primordial tensor background. It does not favour the matter bounce *specifically*: $\Omega_{\text{GW}} \propto f^2$ is the causality-limited infrared slope common to scalar-induced backgrounds of essentially any origin. Nor does this channel constrain f_{NL} : it measures a slope, not a bispectrum amplitude, and is identical at $-35/16$ and $-35/8$. Converting it into a discriminating test requires propagating the matter-bounce scalar spectrum through the induced-GW kernel to predict an *amplitude* in the nHz band; that computation is not attempted here.

TABLE II. Channel II. f_{PBH} at $M = 10^{20}$ g, with σ calibrated at each threshold so that the Gaussian case gives $f_{\text{PBH}} = 1$. Entries “0” mean ζ_c exceeds the ceiling of Eq. (5). Perturbativity $0.6|f_{\text{NL}}|\sigma \leq 0.04$ throughout.

ζ_c	σ_*	$f_{\text{PBH}}(-35/16)$	$f_{\text{PBH}}(-35/8)$	ratio
0.05	0.006325	7.32×10^{-3}	3.75×10^{-6}	1.95×10^3
0.08	0.010120	1.24×10^{-4}	1.12×10^{-14}	1.11×10^{10}
0.10	0.012650	3.75×10^{-6}	0	—
0.12	0.015179	4.29×10^{-8}	0	—
0.15	0.018974	1.82×10^{-12}	0	—

IV. CHANNEL II: PRIMORDIAL BLACK HOLE ABUNDANCE

A. Press–Schechter with local quadratic non-Gaussianity

Writing $\zeta = \zeta_G + A(\zeta_G^2 - \sigma^2)$ with $A = \frac{3}{5}f_{\text{NL}}$ and $\zeta_G \sim \mathcal{N}(0, \sigma^2)$ [8, 9], the mass fraction is $\beta(\zeta_c) = P(\zeta > \zeta_c)$, obtained exactly by inverting the quadratic. Present-day abundance follows the standard radiation-era relation [10]

$$f_{\text{PBH}}(M) = 1.68 \times 10^8 \left(\frac{\gamma_c}{0.2} \right)^{1/2} \left(\frac{g_*}{106.75} \right)^{-1/4} \times \left(\frac{M}{M_\odot} \right)^{-1/2} \beta(M), \quad (4)$$

with $\gamma_c = 0.2$, $g_* = 106.75$, and $M = 10^{20}$ g.

B. An absolute ceiling that doubles

For $f_{\text{NL}} < 0$ the map is a downward parabola, so

$$\zeta_{\text{max}} = -\frac{5}{12f_{\text{NL}}} + \frac{3}{5}|f_{\text{NL}}|\sigma^2, \quad (5)$$

and no realisation exceeds it. The leading term scales as $1/|f_{\text{NL}}|$: $\zeta_{\text{max}} = 0.09524$ at $-35/8$ and 0.19048 at $-35/16$, a ratio of exactly 2. This is the sharpest statement the channel makes about the two candidate values.

C. Abundance at fixed amplitude

The curvature power-spectrum amplitude is set by the source, and f_{NL} then decides whether black holes are overproduced. Calibrating σ on the Gaussian case so that $f_{\text{PBH}} = 1$ and reading off the two matter-bounce values at that same σ gives Table II.

Negative f_{NL} suppresses black hole formation, and $-35/16$ is a substantially weaker suppressor than $-35/8$: three to ten orders of magnitude at fixed amplitude. At $\zeta_c = 0.05$ the halved value lands inside the $10^{-3} \leq f_{\text{PBH}} \leq 1$ band reported for negative- f_{NL} bounce scenarios [11], while $-35/8$ falls three orders below it.

TABLE III. Channel III. Detection significance of $f_{\text{NL}}^{\text{local}} = -2.1875$. Forecast σ values are quoted from the cited abstracts; the significances are $|f_{\text{NL}}|/\sigma$ and $|f_{\text{NL}}|r/\sigma$. MegaMapper is an unapproved design and is quoted only at its published order-unity level; no systematic budget is transferred to it.

Survey	$\sigma_{f_{\text{NL}}}$	bare	r -projected
SPHEREx, bispectrum only [13]	0.7	3.13σ	2.63σ
SPHEREx, $P + B$ target [13, 14]	0.5	4.38σ	3.68σ
MegaMapper class [15, 16]	~ 1	2.19σ	1.84σ

D. Limitation of the quadratic truncation

At the standard curvature thresholds $\zeta_c \approx 0.45$ –1, the ceiling of Eq. (5) forces $\beta = 0$ identically for *both* negative values in the rare-tail regime. This is a property of the truncated quadratic map, not a physical statement, and it is why the compaction-function formation criterion is used in the literature [11]. Choudhury *et al.* treat $f_{\text{NL}} = (-39.95, -35/8)$ in an effective-field-theory bounce with an ultra-slow-roll phase and report $10^{-3} \leq f_{\text{PBH}} \leq 1$, complete mitigation of overproduction, and a perturbativity bound $|f_{\text{NL}}| \lesssim 60$; they do not treat $-35/16$. Redoing that calculation at $-35/16$ is the principal open item of this work and is *not* claimed here.

V. CHANNEL III: THE LARGE-SCALE-STRUCTURE BISPECTRUM

A. Current constraints

The DESI DR1 analysis of 1,631,716 luminous red galaxies ($0.6 < z < 1.1$) and 1,189,129 quasars ($0.8 < z < 3.1$) [12] gives $f_{\text{NL}}^{\text{local}} = -3.6_{-9.1}^{+9.0}$ at 68% confidence with a merger model for the quasar response, or $f_{\text{NL}}^{\text{local}} = +3.5_{-7.4}^{+10.7}$ assuming universality. The prediction of Eq. (1) sits 0.16σ from the former and 0.77σ from the latter. The discriminating power is $|f_{\text{NL}}|/\sigma = 0.24$: DESI DR1 is consistent with the prediction and cannot test it.

B. Forecast reach

The matter-bounce bispectrum is not exactly the local shape; we adopt the noise-weighted overlap $r = 0.84$, so an experiment quoting σ_{local} constrains the bounce amplitude at σ_{local}/r . Table III gives both.

At $-35/8$ the same rows read 6.25σ , 8.75σ and 4.38σ bare. This is the only channel in which the factor of two is observationally consequential.

VI. DISCUSSION

A. The joint statement

The three channels are mutually consistent at $f_{\text{NL}}^{\text{local}} = -35/16$. The pulsar timing slope agrees at 1.14σ and re-

jects the two competing spectral shapes; the primordial black hole channel is suppressed but not extinguished, and lands in the sizeable-abundance band; the bispectrum prediction is consistent with the best current measurement and reachable by SPHEREx. No channel is in tension with another.

B. Where the consistency is weak

Two of the three channels are insensitive to the parameter. The pulsar-timing result is a statement about a slope that any scalar-induced background shares, and it is identical at both candidate f_{NL} values. The black hole channel is reported only at Press-Schechter level. The operative uncertainty is therefore internal: only the bispectrum channel separates $-35/16$ from $-35/8$, and it separates them by more than 3σ at SPHEREx sensitivity, so settling the factor of two by an independent derivation is a prerequisite for a sharp prediction rather than a bookkeeping detail.

C. Next steps

(i) a compaction-function recomputation of Channel II at $-35/16$; (ii) an independent gradient-expansion or δN derivation of Eq. (1); (iii) propagation of the matter-bounce scalar spectrum through the induced-GW kernel to predict the nHz *amplitude*, converting Channel I from a consistency check into a test; (iv) re-derivation of the shape overlap r at the $-35/16$ fiducial.

DATA AND CODE AVAILABILITY

All three channel computations are committed at `research/track_a3_multichannel/pta_gamma_reproduce.py`, `pbh_abundance_fnl.py`, `survey_reach_fnl.py`, with results in `outputs/` and reproducibility manifests at `reproducibility/manifests/experiments/a3-*.json`. The pulsar-timing chain is `pipelines/p3_pta_mcmc/free_spectrum_real_2026-05-01/chain_real_freespec.npy` (SHA-256 50abc38a...10fc), derived from the NANOGrav 15-yr KDE free spectra, Zenodo 10.5281/zenodo.8060824. Every computation ran on a local CPU in under 0.1 s at zero compute cost.

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